

Before class

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A 15-minute warm-up for the lecture. Make one prediction, read six short cards, then try four quick calculations. Tick one answer for each multiple-choice question. Check yourself with the answer key on the last page.

By the end of the lecture you should be able to...

- EM-LO1** Define energy and heat, and convert between joules, calories and nutritional Calories
- EM-LO2** Convert between temperature scales and calculate heat gained or lost using $q = mc\Delta T$
- EM-LO3** Describe phase changes and calculate the energy involved using heats of fusion and vaporisation
- EM-LO4** Interpret the heating curve of water
- EM-LO5** Define bond energy, estimate ΔH from bond energies, and classify reactions as endothermic or exothermic
- EM-LO6** Relate energy changes to biochemical reactions such as glucose oxidation and ATP hydrolysis

1 · Predict (not scored)

You squeeze an instant cold pack for a patient's sprained ankle. Inside, ammonium nitrate dissolves in water and the pack turns cold within seconds. Which way does the heat flow?

- ☐ A From the pack into the ankle
- ☐ B From the ankle into the pack
- ☐ C No heat flows; the pack makes 'cold'
- ☐ D Heat is destroyed inside the pack

2 · Read and check

EM-LO1 Ball 7.0–7.1

Energy, heat and their units

Energy is the ability to do work. **Heat** is the *transfer* of energy between objects that are at different temperatures. An object contains energy; heat is what flows.

The SI unit of energy is the **joule (J)**; $1 \text{ kJ} = 1000 \text{ J}$. The **calorie (cal)** is an older unit: $1 \text{ cal} = 4.184 \text{ J}$.

Food labels use the nutritional **Calorie** with a capital C: $1 \text{ Cal} = 1 \text{ kcal} = 1000 \text{ cal} = 4.184 \text{ kJ}$. Malaysian labels show both kcal and kJ, so you can check the factor on any snack packet.

Q1. A food label says one serving provides 110 Calories. How many calories (small c) is that?

- ☐ A 0.110 cal
- ☐ B 110 cal
- ☐ C $1.10 \times 10^5 \text{ cal}$
- ☐ D $4.60 \times 10^5 \text{ cal}$

Heat and temperature

Temperature measures the **average kinetic energy** of the particles. It is measured in °C, °F or K, where $K = ^\circ\text{C} + 273.15$. Normal body temperature is 37.0 °C.

Heat flows from a **hotter** object to a **colder** one. The heat needed to change a temperature is $q = m \times c \times \Delta T$, where $\Delta T = T_{\text{final}} - T_{\text{initial}}$.

The **specific heat (c)** is the energy to warm 1 g of a substance by 1 °C. Water's is high, 4.184 J/g·°C, so water warms and cools slowly. That is why warming a litre of IV fluid takes a lot of energy.

Q2. Which statement is correct?

- | | |
|--|---|
| ☐ A Heat and temperature mean the same thing | ☐ B Temperature measures the average kinetic energy of the particles |
| ☐ C Heat flows from a colder object to a hotter one | ☐ D Water has a low specific heat, so it heats up quickly |

Phase changes

Matter is solid, liquid or gas depending on how much energy its particles have.

Melting, vaporisation and sublimation absorb heat (**endothermic**). **Freezing, condensation and deposition** release heat (**exothermic**).

While a phase change happens, the temperature stays **constant**. The heat is found with $q = m \times \Delta H_{\text{fus}}$ (melting) or $q = m \times \Delta H_{\text{vap}}$ (boiling). For water, $\Delta H_{\text{fus}} = 334 \text{ J/g}$ and $\Delta H_{\text{vap}} = 2260 \text{ J/g}$.

Q3. Which change of state releases heat to the surroundings?

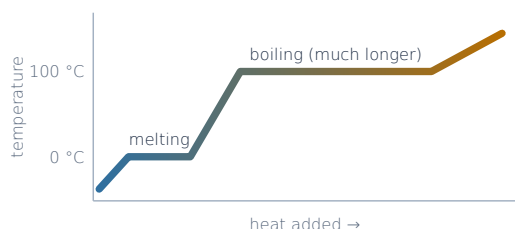
- | | |
|--|---|
| ☐ A Melting | ☐ B Vaporisation |
| ☐ C Sublimation | ☐ D Condensation |

The heating curve of water

A heating curve plots temperature against heat added as ice is warmed all the way to steam.

The **slopes** are one phase warming up (use $q = mc\Delta T$). The **flat sections** are phase changes at constant temperature (use $q = m\Delta H$): melting at 0 °C and boiling at 100 °C.

The boiling section is about 7 times longer than the melting section, because ΔH_{vap} is much bigger than ΔH_{fus} .



Q4. On the heating curve of water, what is happening along the flat section at 100 °C?

- | | |
|--|---|
| ☐ A Steam is warming up | ☐ B Water is boiling at constant temperature |
| ☐ C Ice is melting | ☐ D No heat is being added |

Bond energy and reactions

Atoms bond because bonded atoms have **lower** energy. **Breaking** a bond always **needs** energy; **forming** a bond always **releases** energy. This amount is the **bond energy**.

A reaction's energy change can be estimated as **$\Delta H \approx \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$** .

Exothermic: heat is a product, ΔH is negative, the surroundings warm up. **Endothermic**: heat is a reactant, ΔH is positive, the surroundings get cold.

Q5. Burning methane in a Bunsen burner has $\Delta H = -890$ kJ per mole. Which statement is correct?

- | | |
|---|---|
| ☐ A It is endothermic and the surroundings get cold | ☐ B It is exothermic and releases heat to the surroundings |
| ☐ C Its products are higher in energy than its reactants | ☐ D Breaking the bonds in methane releases energy |

EM-LO6 Ball 7.5

Energy in the body

The body gets energy by oxidising glucose: $\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 \rightarrow 6 \text{CO}_2 + 6 \text{H}_2\text{O} + \text{about } 2800 \text{ kJ (670 kcal) per mole.}$

Carbohydrates and proteins supply about **17 kJ/g (4 kcal/g)**; fats supply about **38 kJ/g (9 kcal/g)**.

Cells store and spend energy through **ATP**. Converting ATP to ADP releases about **31 kJ/mol (7.5 kcal/mol)**: small, but enough to drive other reactions in the cell.

Q6. Which nutrient supplies the most energy per gram?

☐ A Carbohydrate

☐ B Protein

☐ C Fat

☐ D Water

3 · Maths warm-up

EM-LO1

Q7. A snack provides 2.50 kcal per gram. Convert this to kJ per gram. (1 kcal = 4.184 kJ)

Working:

Answer: _____ kJ/g

EM-LO2

Q8. Insulin must be stored in a fridge at no more than 8 °C. Express 8 °C in kelvin.

Working:

Answer: _____ K

EM-LO2

Q9. A bag of IV fluid cools from 37.0 °C to 25.0 °C. What is ΔT ?

Working:

Answer: _____ °C

EM-LO2

Q10. Rearrange $q = m \times c \times \Delta T$ to find the specific heat, c .

☐ A $c = q \times m \times \Delta T$

☐ B $c = q \div (m \times \Delta T)$

☐ C $c = (m \times \Delta T) \div q$

☐ D $c = q - m - \Delta T$

4 · Your results

Mark your answers with the key, then fill in the table. Any outcome that is not full marks shows which part of the lecture to watch closely.

Outcome	Questions	Lecture part	My score
EM-LO1	Q1, Q7	Part 1	/ 2
EM-LO2	Q2, Q8, Q9, Q10	Part 2	/ 4
EM-LO3	Q3	Part 3	/ 1
EM-LO4	Q4	Part 3	/ 1
EM-LO5	Q5	Part 4	/ 1
EM-LO6	Q6	Part 5	/ 1
Total			/ 10

Answer key

Q	Outcome	Answer	Why
Predict	EM-LO5	B	Heat always flows from hotter to colder. The dissolving process absorbs heat (it is endothermic), so heat flows out of the ankle and into the pack. The ankle loses energy, so it feels cold. There is no such thing as 'cold' flowing; cold is just less energy. You will meet endothermic and exothermic processes in Part 4.
Q1	EM-LO1	C	Capital-C Calorie = 1 kcal = 1000 cal. So $110 \text{ Cal} \times 1000 \text{ cal/Cal} = 110\,000 \text{ cal} = 1.10 \times 10^5 \text{ cal}$. (4.60×10^5 is the value in joules, not calories.)
Q2	EM-LO2	B	Temperature is the average kinetic energy of the particles; heat is energy that flows because of a temperature difference, always from hotter to colder. Water's specific heat is high, not low.
Q3	EM-LO3	D	Condensation (gas $\rightarrow$ liquid) is exothermic: the particles lose energy, and it goes to the surroundings. This is why a steam burn is worse than a hot-water burn. The other three absorb heat.
Q4	EM-LO4	B	At 100 °C, liquid water turns into steam. Heat is still being added, but it goes into changing the state, not into raising the temperature.
Q5	EM-LO5	B	A negative ΔH means heat is released: exothermic. The products are lower in energy than the reactants. Breaking bonds always costs energy; the energy is released when the new bonds form.
Q6	EM-LO6	C	Fat supplies about 38 kJ/g (9 kcal/g), more than twice the 17 kJ/g (4 kcal/g) of carbohydrate or protein. Water supplies no energy.
Q7	EM-LO1	10.5 kJ/g	$2.50 \text{ kcal/g} \times 4.184 \text{ kJ/kcal} = 10.46 \text{ kJ/g} = \mathbf{10.5 \text{ kJ/g}}$ (3 significant figures).
Q8	EM-LO2	281 K	$K = ^\circ\text{C} + 273.15 = 8 + 273.15 = 281.15$, reported as 281 K .
Q9	EM-LO2	-12 °C	$\Delta T = T_{\text{final}} - T_{\text{initial}} = 25.0 - 37.0 = \mathbf{-12.0 \text{ }^\circ\text{C}}$. The minus sign tells you the fluid lost heat.
Q10	EM-LO2	B	Divide both sides by $(m \times \Delta T)$: $c = q \div (m \times \Delta T)$. You will use this in Part 2 to identify an unknown metal.

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